

MIT 805: Big Data Semester Project (2026): Large-Scale Data Analysis and Distributed Processing with PySpark

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LARGE-SCALE ANALYSIS OF NYC HVFHV TRIPS

1. Data, access and suitability

The dataset we used is the official New York City Taxi and Limousine Commission (TLC) High Volume For-Hire Vehicle (HVFHV) Trip Record.

collection: Each row represents one dispatched trip and records provider licence, request/pickup/drop-off timestamps, taxi zones, distance, duration, fares, fees, tips, driver pay and service flags. NYC Open Data describes City data as free public data. The files are directly downloadable without registration, and the use is governed by the NYC government Terms of Use. The TLC page states no separate dataset-specific licence, so the report cites both the source and governing terms. No passenger names or direct personal identifiers appear, making the records suitable for academic aggregate analysis.

2. Scale and characteristics

The selected collection contains 60 complete monthly Parquet files from January 2021 to December 2025. The files total 27.55 GB, satisfying the 25 to 40 GB raw-data requirement. The same full 27.55 GB collection is the working dataset and PySpark input. No pre-analysis sampling was applied, so both exceed the 12 GB and 3 GB thresholds. PySpark processed 1,102,562,887 records and 25 source fields. Data types include strings, timestamps, integer zone identifiers, continuous financial and distance values, and binary service flags.

3. Preparation and data quality

All 60 expected months were present. Schema evolution required explicit preparation: pickup and drop-off zone identifiers alternated between INT and BIGINT and were standardised to long before joining the monthly files. The CBD congestion-fee field appears only in the 12 files from 2025, consistent with TLC documentation that the field was added for 2025 data. Congestion-fee nulls in the files before 2025 were therefore treated as absence instead of zero. The important trip variables were highly complete: only five request timestamps were null, while pickup/drop-off timestamps, zone IDs, distance, duration and main financial fields had no nulls. Originating_base_num (27.19%) and on_scene_datetime (22.06%) were treated as optional fields. Logical checks identified pickup-before-request records (1.0288%), negative base fares (0.0485%), non-positive distance (0.0181%), drop-off not after pickup (0.0043%),

negative driver pay (0.0018%) and non-positive duration (0.0001%). Timeline, distance and zone-valid records form the core analysis population (98.95%).

4. Exploratory findings

Data Collection & Analysis: Monthly volume increased from 11.91 million trips in January 2021 to 22.11 million in December 2025 (+85.65%). Annual trips rose from 174.60 million in 2021 to 243.59 million in 2025, although annual growth slowed from 21.66% in 2022 to 1.72% in 2025. Detailed 2025 analysis found the lowest hourly demand at 03:00 (3.46 million trips), a morning local peak at 08:00 (12.23 million), and the overall peak at 18:00 (14.21 million). Saturday was the busiest (39.67 million) and Monday the quietest (29.88 million). Using TLC licence mappings, Uber (HV0003) represented 72.4% of valid 2025 trips and Lyft (HV0005) 27.6%.

5. Evidence for seven Big Data Vs

Seven V's	Evidence in the NYC HVFHV trips data
Volume	27.55 GB and 1.103 billion rows required distributed processing.
Velocity	TLC publishes monthly, and monthly demand reached 22.11 million by December 2025.
Variety	25 fields span timestamps, identifiers, continuous values and categorical flags.
Veracity	Source disclaimers, schema drift, structural nulls and temporal inconsistencies require validation.
Variability	Pronounced monthly, hourly and weekday changes coexist with slowing annual growth.
Visualisation	Monthly, hourly and weekday charts convert billion-row aggregates into interpretable patterns.
Value	The results support demand forecasting, fleet positioning, driver scheduling, congestion analysis and policy evaluation.

6. Business and societal value and limitations

Demand peaks can guide vehicle supply and reduce passenger waiting, while zone vs time analysis can support congestion management and transport integration. Public agencies can monitor mobility recovery and evaluate fees or accessibility programmes. Providers can schedule drivers and allocate incentives more efficiently. This supports operational improvement. TLC states that submissions are provider-generated, and it does not guarantee accuracy or completeness. Provider concentration, changing schemas, non-random missingness, lack of a unique trip ID, and the 2025-only detailed temporal analysis limit generalisation.

GitHub link: [Sav-lets/nyc-hvfhv-big-data-analysis-mit805-project](https://github.com/Sav-lets/nyc-hvfhv-big-data-analysis-mit805-project)

References

- [1] NYC TLC. TLC Trip Record Data. <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page> (accessed 29 Aug 2026).
- [2] NYC TLC. Data Dictionary - High Volume FHV Trip Records, 18 Mar 2025. https://www.nyc.gov/assets/tlc/downloads/pdf/data_dictionary_trip_records_hvfhs.pdf
- [3] City of New York. NYC.gov Terms of Use. <https://www.nyc.gov/main/terms-of-use> (accessed 29 Aug 2026)

Appendix A - Supporting visualisations

Data Summary:

Raw, Working and Processed Data Size	Monthly files volume	Trip Records	Identified Valid Records
27.55GB	60	1.103 billion	98.95%

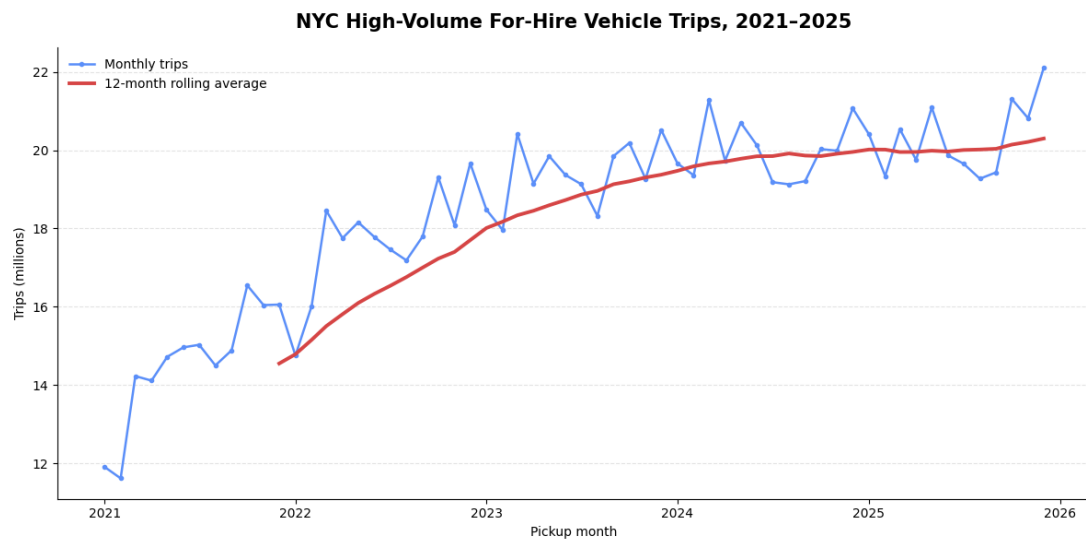


Figure 1: Monthly HVFHV trip volume and 12-month rolling average, 2021-2025.



Figure A1. Valid 2025 HVFHV trips by pickup hour.

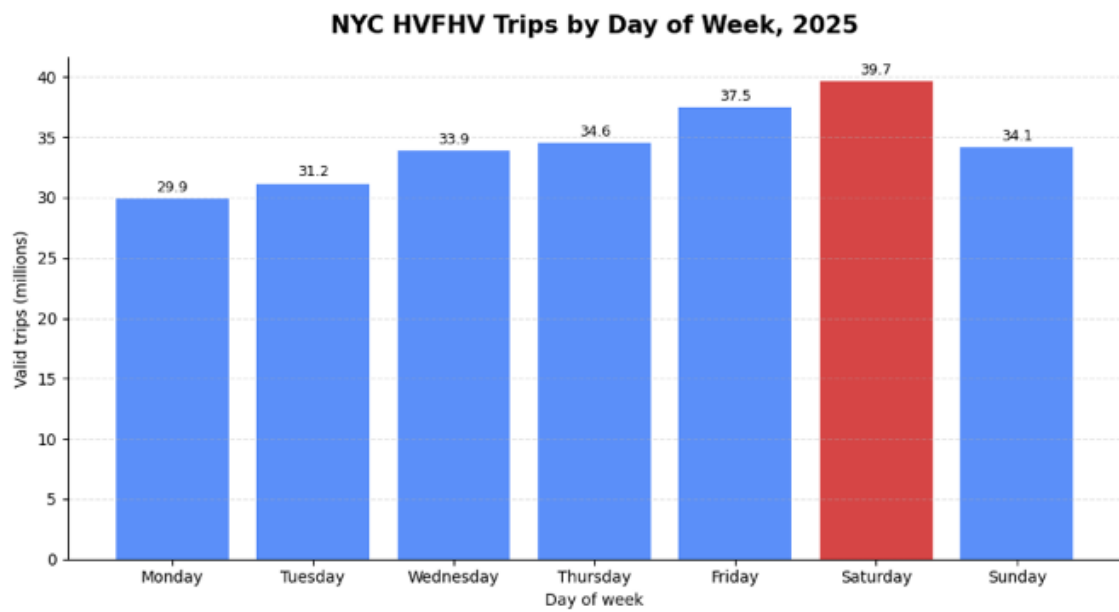


Figure A2. Valid 2025 HVFHV trips by day of week.

Quality Issues:

Quality issue	Records	Percent
Pickup before request	11,342,990	1.0288%
Negative base fare	534,792	0.0485%
Nonpositive trip miles	199,558	0.0181%
Drop-off not after pickup	47,168	0.0043%
Negative driver pay	19,578	0.0018%
Nonpositive trip time	1,049	0.0001%
Invalid pickup/drop-off zone	0	0.0000%